Rahul Jaikrishna

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Computer Science student focused on formal methods, AI/ML, and trustworthy intelligent systems, with a research interest in agentic assurance and reliable AI. I love to explore how formal verification, program analysis, and machine learning can be combined to build AI systems that are robust and verifiable.

PROFESSIONAL EXPERIENCE

Student Researcher, University of California Berkeley

Berkeley, California, USA (on site 3 months) then Remote — May 2025 - Present

- Collaborating with Prof. Christopher Fletcher and Dr. Sushant Dinesh on formal verification of hardware to mitigate speculative execution vulnerabilities
- Developed formal models of microarchitectural behavior to prove security guarantees against side-channel and transient execution attacks.
- Applying verification techniques to open-source processors (BOOM) using Chisel and Chipyard to analyze and reason about speculative state leakage.

Software Enginnering Intern, The College App

Illinois (Remote) — Sep 2024 – Feb 2025

- Built and deployed a production-grade AI college counseling assistant used by real students navigating the U.S. admissions process.
- Architected and implemented a RAG pipeline combining vector search and LLM reasoning for context-grounded responses.
- Designed embedding workflows, retrieval logic, and prompt orchestration to improve factual accuracy and reduce hallucinations.

AI4Y Trainee, Intel Corporation

Bangalore, India — May 2019 – Sep 2020

- Selected among the Top 100 students nationwide for a competitive, multi-month AI training initiative.
- Built Cyber Detective, an NLP-based system designed to detect and combat online hate speech.
- Developed end-to-end pipeline including text preprocessing, model training, and classification for harmful content detection.

PAPERS & RESEARCH

Counterexample Guided Defense Synthesis (submitted @ SPACE 2026)

- Achieved certified detection of adversarial inputs with by synthesizing formally verified drift based detectors using counterexample guided optimization and neural network verification.

Cross Paradigm Defense for Vision-Language Models (submitted @ IEEE TPS 2026)

- Improved resilience against adaptive white box attacks by using cryptographically keyed selection of vision encoders and VLMs with multi-stage embedding drift and semantic consensus detection.

Probe Guided Active Intent Elicitation for Software Engineering (to be submitted to SANER 2027)

- Uses LLM activation probes to measure prompt uncertainty and select the minimum set of clarifying questions that maximizes ambiguity reduction.

PROJECTS

Self-Hosted DNS Resolver

and 0.11% false-positive rate in production deployment with high-performance blocklist engine and ML-based filtering.

Cyber Detective (Patent Awaited)

AWARDS

2x Winner from India at Intel® AI Global Impact Festival

Venture Lab Series Winner

EDUCATION

B.Tech, Computer Science
PES University
2023-2027

- Lead, AI/ML — Encode.AI
- CGPA: 8.95

DPS East, Bangalore
March 2023-2023

- 12th: 93%
- 10th: 91%
- School Distinction Award for subject topper in AI

PATENT

"System and Method for Tracking and Detecting Cyber-Bullies"
Application No: 202141030788
Status: Granted

SKILLS

- **Languages:** Python, C, Assembly, Bash
- **Systems:** Bare-metal kernels, Memory management, Linux & DNS internals
- **ML & AI:** ML/DL, NLP, RAG, CNNs, Model optimization
- **Cloud & Infra:** AWS, DoH, Production systems
- **Tools:** PyTorch/TensorFlow, FAISS, Git, Docker